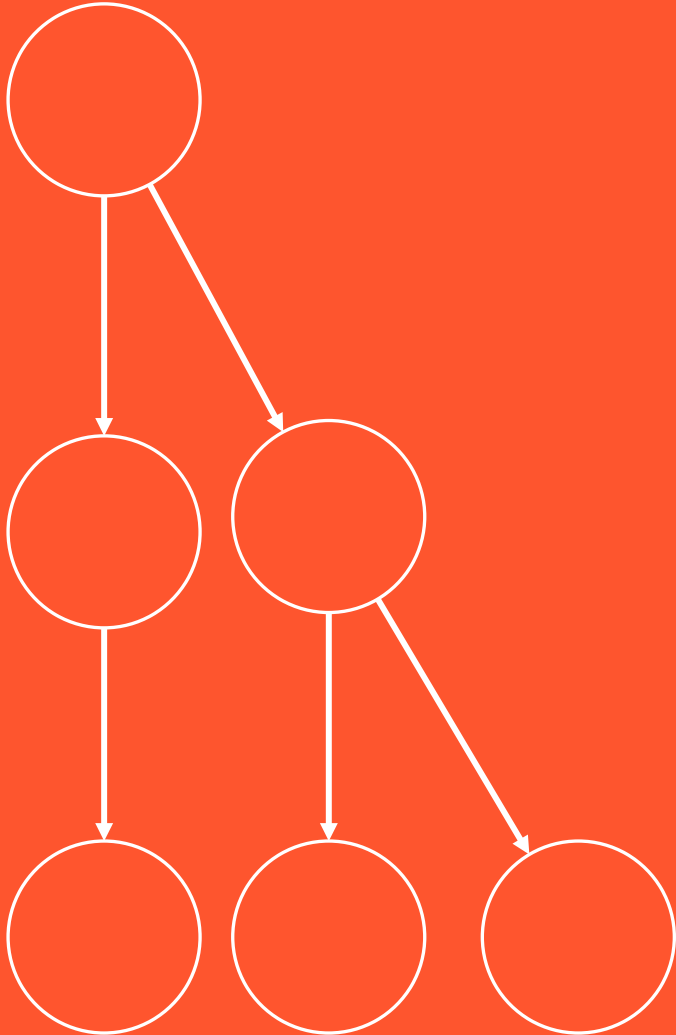


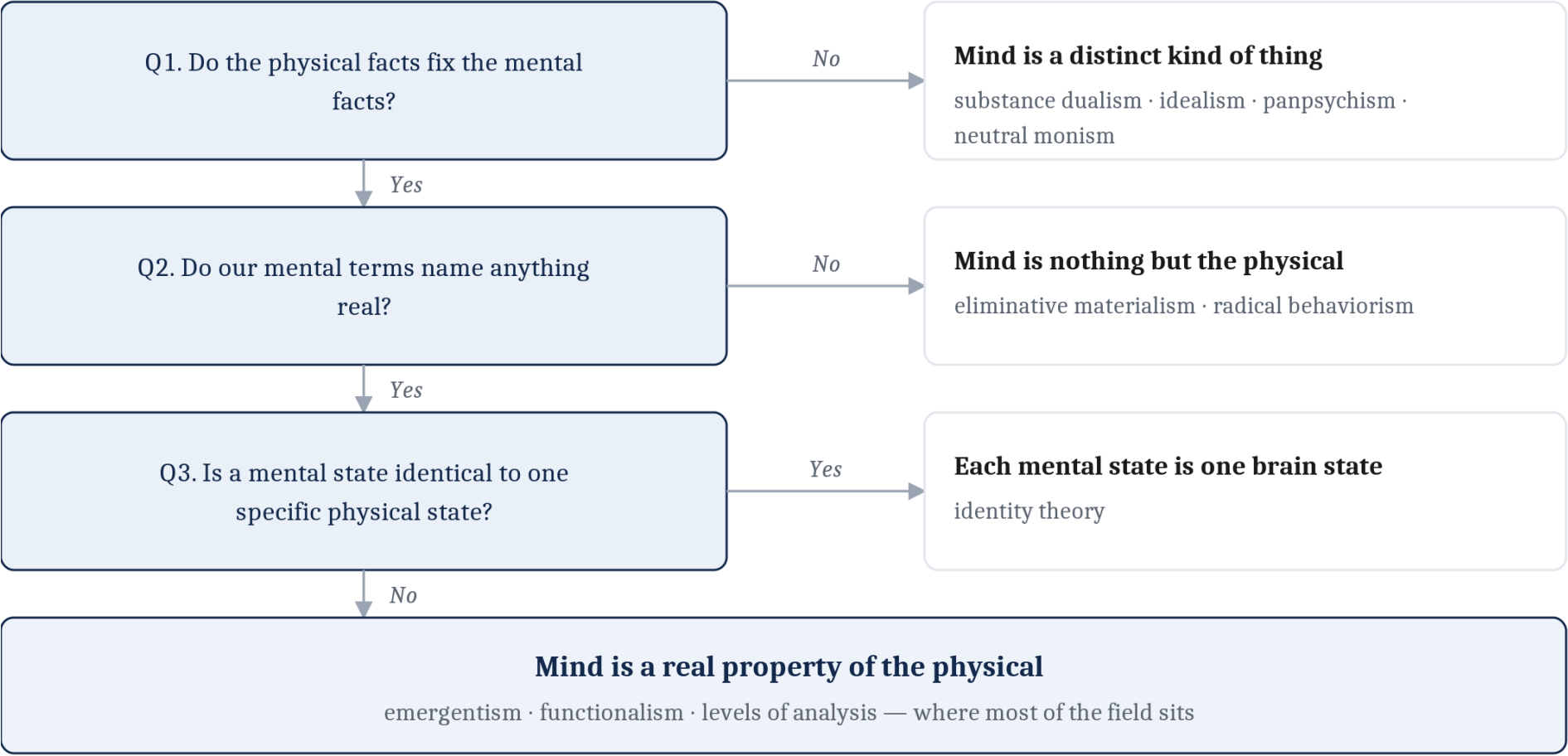
BCOG 100

1.2 Levels of Analysis

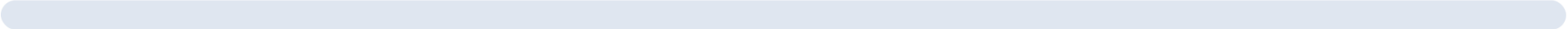


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Where We Stopped, and What I Owe You



Q4. How freely can such a mind be studied in its own terms?
a matter of degree, not another fork — and the field is genuinely split along it



Strong autonomy: psychology as chemistry is to physics

Realization-sensitive: embodiment, the speed of neurons

Start Where There Is No Mind At All



A **leaf**. Ask three different questions about it:

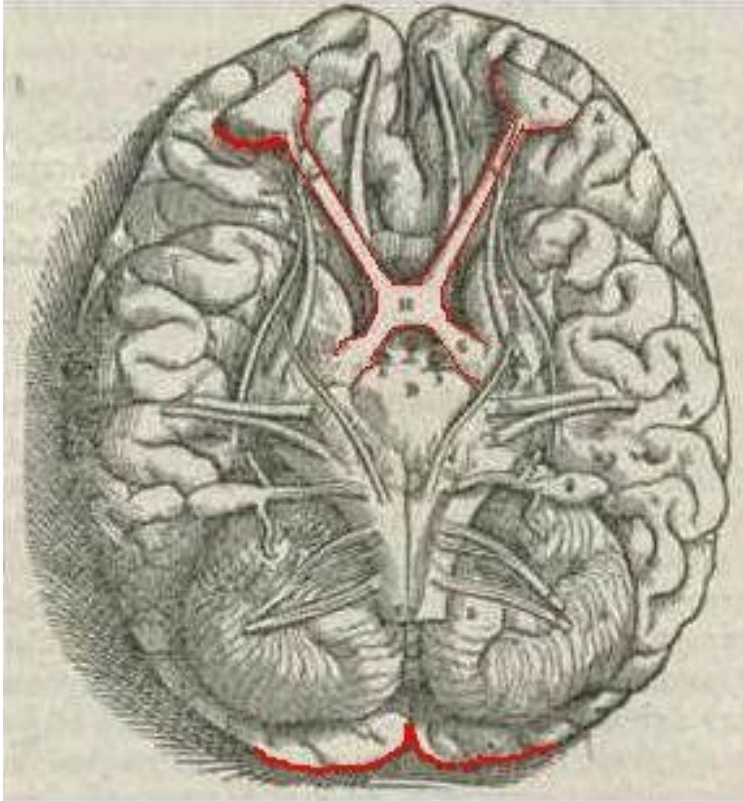
What is it for? Capturing energy from sunlight.

By what strategy? Spread thin and wide to catch as much light as possible; track the sun; run a chemical cascade that stores the energy in a bond.

What is it made of? Chloroplasts, stacked membranes, pigment molecules, pores that open and close.

Now: **which of those three is the real explanation?**

Where This Framework Came From



1970s. Neuroscientists are learning an enormous amount about individual neurons in the visual system — and getting no closer to explaining how **seeing** works.

David Marr's complaint: trying to understand vision by studying neurons alone is like trying to understand the flight of a bird by studying only feathers.

It cannot be done.

Not because feathers are irrelevant. Because the question you are asking is not a question about feathers.

Marr's Three Levels

Computational

What problem is being solved, and why?

Algorithmic

By what strategy?

Implementational

In what physical stuff, and how?

Marr's contribution was not discovering that these differ. It was naming them, sharpening them, and insisting **all three are needed**.

One Problem, Many Solutions

The problem: put a shelf of books in alphabetical order.

Stated without any procedure at all. It is a fact about the *task*.

The procedures. Take one book at a time and slot it where it belongs. Or split the shelf in half, sort each half, merge them. Or find the earliest, put it first, repeat.

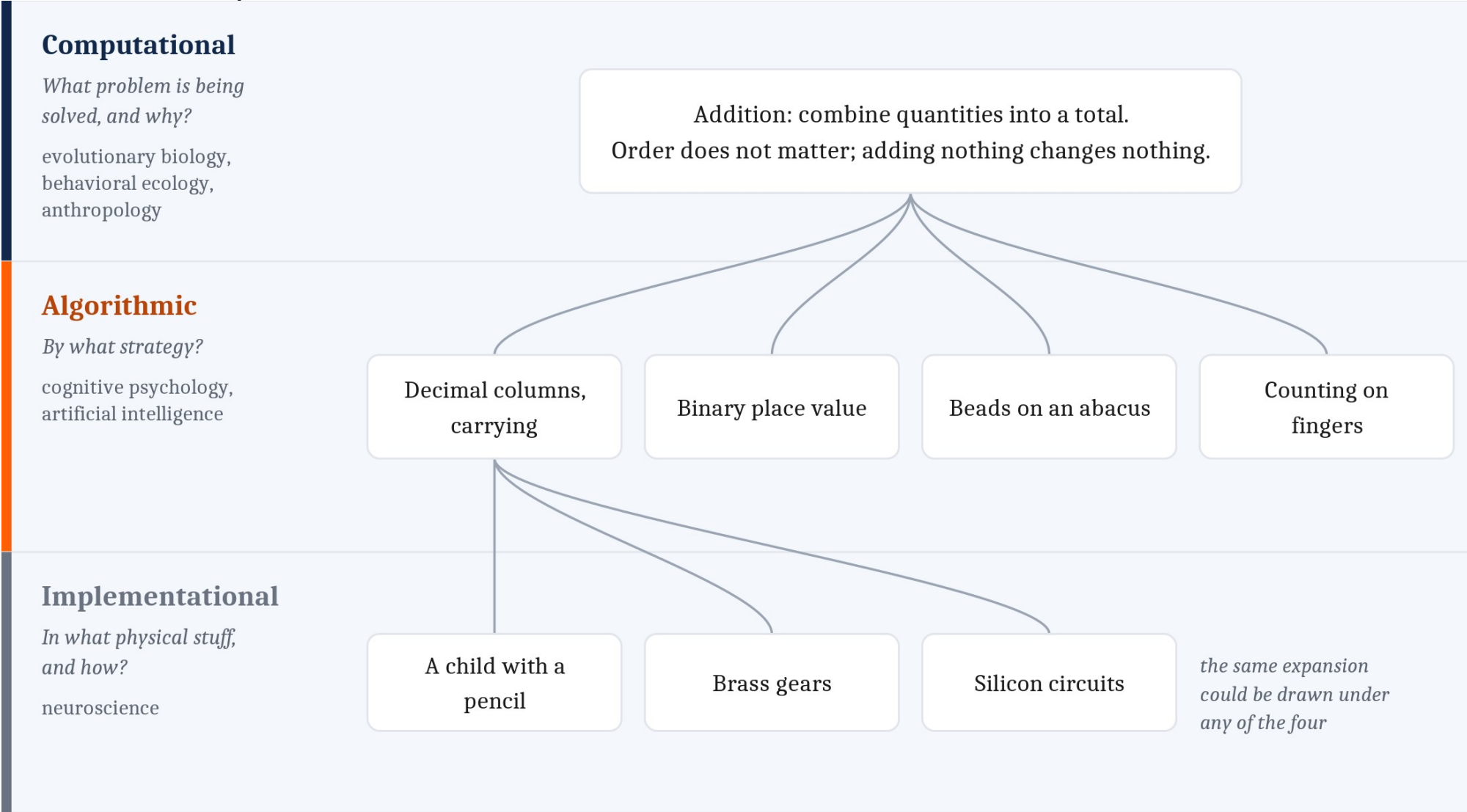
All different. All correct.



One Problem, Many Solutions

The machinery. You, on a Saturday. A library cart and three volunteers. A database, in four milliseconds.

Same procedure. Utterly different stuff.



Why You Cannot Skip One

The relations are **many-to-one in both directions**:

one problem → many algorithms

one algorithm → many implementations

So, fixing one level leaves the others open.

Knowing the problem does not tell you the procedure.

Knowing the procedure does not tell you the hardware.

Which means an answer at one level **cannot stand in for** an answer at another.

These are not three descriptions of one fact. They are answers to three different questions.

Marr went further, and this part **is contested**: he argued the levels are **ordered**, computational first, so analysis should run **top-down**.

We will come back and challenge it at the end.

Why These Three Are the Spine of This Course

The levels line up with how brain and cognitive science **already divides itself**:

Computational — evolutionary biology, behavioral ecology, anthropology

Algorithmic — cognitive psychology, artificial intelligence

Implementational — neuroscience

That is not a coincidence. **The fields grew up around different questions.**

Which explains a great deal about why they talk past one another — and means their findings can be **combined** rather than **ranked**.

Where Does a "Why" Come From With No Designer?

For a calculator: **somebody wanted one.**

A leaf has no designer. Neither does an eye, or a memory system.

For living things, the computational question becomes one about **ancestry and survival** — the **adaptive why.**

But an adaptive why is a **hypothesis to be tested, not a story to be assumed.**

A designer's intent	Natural selection	A cultural practice
A calculator does addition because its makers wanted a tool for arithmetic.	An eye recovers the world because ancestors who saw it survived and reproduced.	Reading and handling money are goals set by a practice and handed down socially.
for an artifact	for a biological trait — the adaptive why	for a learned human skill
A human mind draws on all three at once.		

The Third Source: Culture

Many problems a modern mind solves were set by **cultural practice**. Natural selection has not had nearly enough time.

Music. Every known human culture makes it. But which scales, which rhythms, which instruments — are handed down, not inherited. And it runs on machinery that was doing other jobs first: hearing out a voice in a noisy room, timing a movement.

Is the capacity for music itself an adaptation, or did culture recruit it? People genuinely disagree — which is what a hypothesis looks like.

	Adapted across generations	Adapted within a lifetime
Timescale	many generations	days to years
Mechanism	natural selection on a lineage	learning and practice
What changes	gene frequencies in a population	one brain
What is passed on	genes	skills, taught

A Worked Case: Vision

The obvious answer: vision is for seeing what is out there.

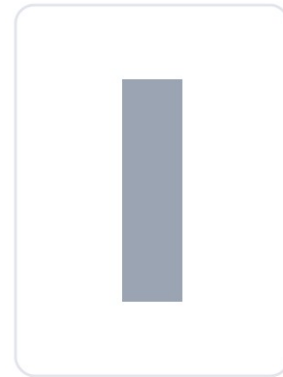
Not wrong. But...

The retinal input is, in effect, **flat**.

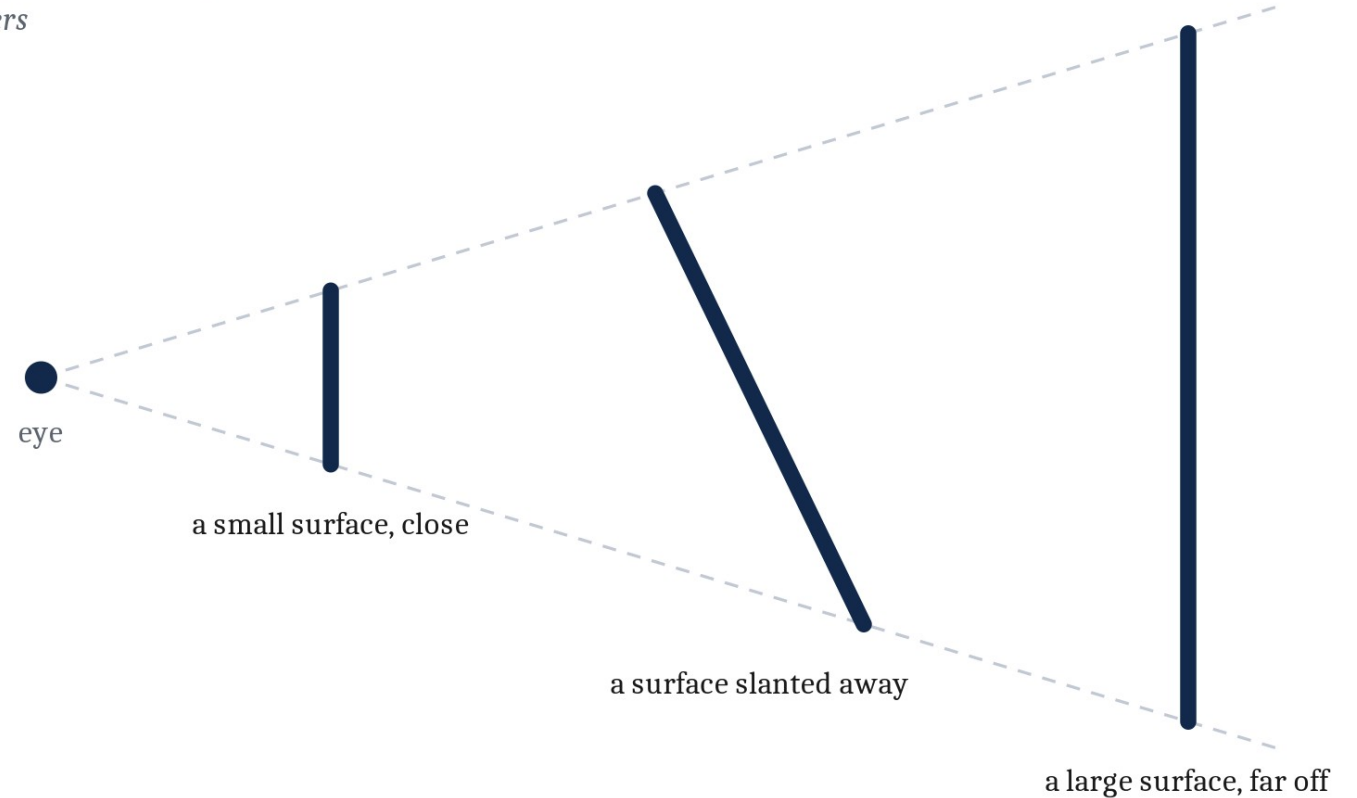
What the animal needs to know about is the **solid world that produced it**.

Infinitely many 3D scenes cast exactly the same flat image. The data alone cannot say which one is out there. This is an **ill-posed problem**.

*every one of them casts exactly the same image —
and so do infinitely many others*

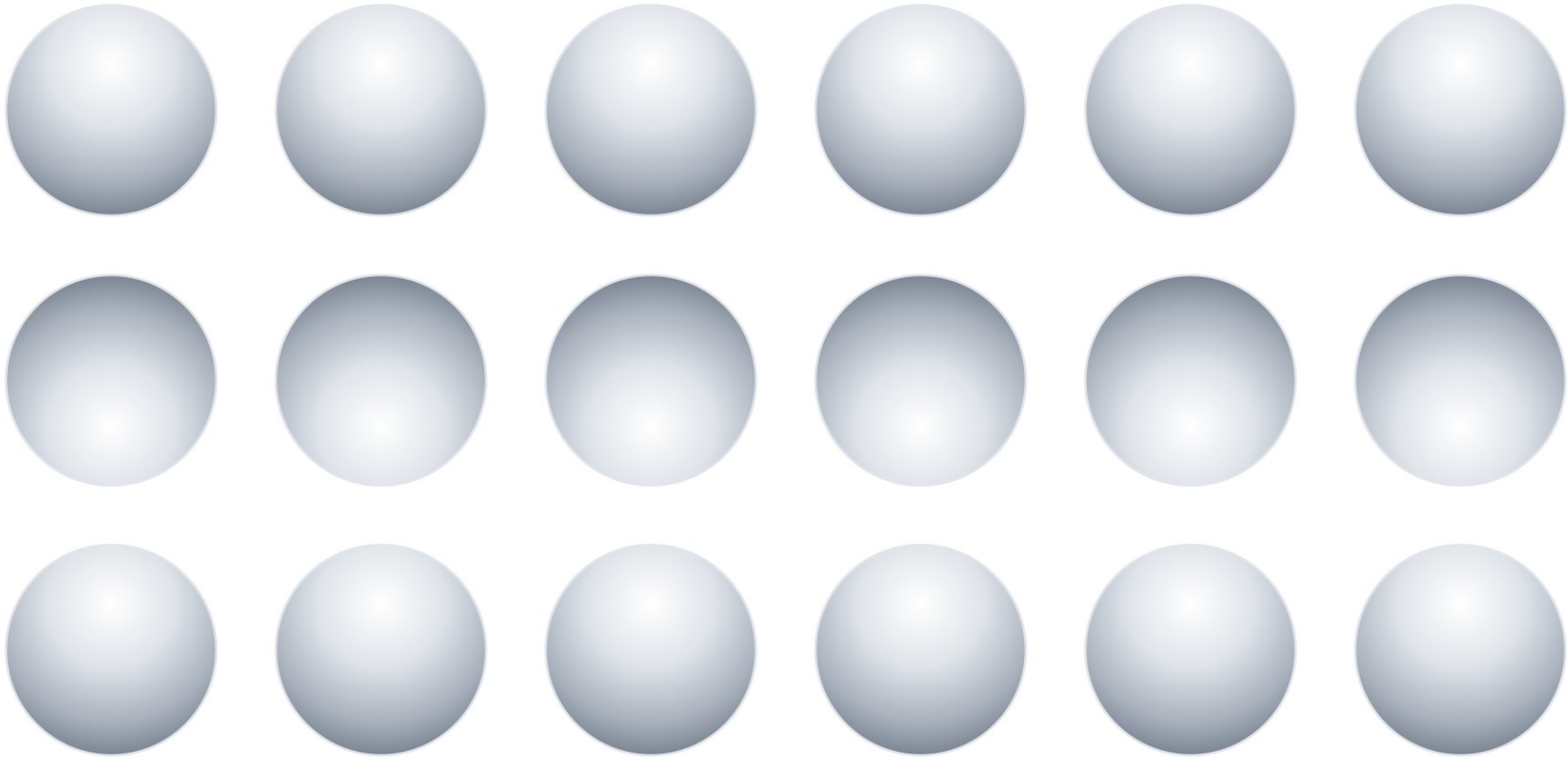


what the eye
receives



Your Visual System Is Guessing Right Now

Which of these are **bumps**, and which are **dimples**?



Vision, All Three Levels

Vision at Three Levels

Computational

What problem is being solved, and why?

Recover the three-dimensional world from a flat image, which the light alone cannot settle.

Algorithmic

By what strategy?

Find sharp brightness changes, recover depth from disparity and texture, group fragments into surfaces, match against things seen before.

Implementational

In what physical stuff, and how?

Retina, then a relay station deep in the brain, then primary visual cortex, then areas tuned to progressively more complex things.

Separable, But Not Independent

Worked through like that, the levels look almost too clean. But they are not sealed rooms.

Computational

the problem's structure narrows
which algorithms are plausible

the representations available
bound which problems can be solved

Algorithmic

an algorithm that stores associations
demands connections that can change

slow neurons force algorithms that
compute many things in parallel

Implementational



And discovery often runs upward: which problem a system solves is itself something we work out by studying how it works. That is a point about the science, not about the systems.

What the Hardware Forces

The clearest case of an upward constraint. A brain cannot win by doing steps *quickly* — so it does not use algorithms that need long sequences. It uses ones that do enormous numbers of things **in parallel**.

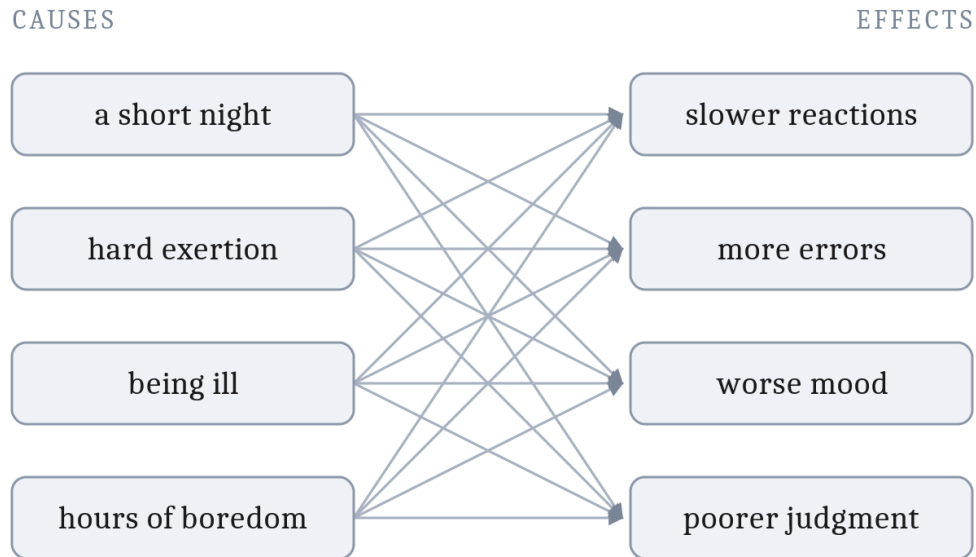
	Neuron	Silicon transistor
Basic signaling operation	roughly a millisecond	far less than a billionth of a second
Speed gap	—	at least a million times faster
Connections per unit	thousands	a handful
What that forces on the algorithm	compute many things at once	compute fast, one step at a time

When Does a Higher-Level Concept Earn Its Keep?

Fatigue is not a thing you can point to in the brain. So why keep the word?

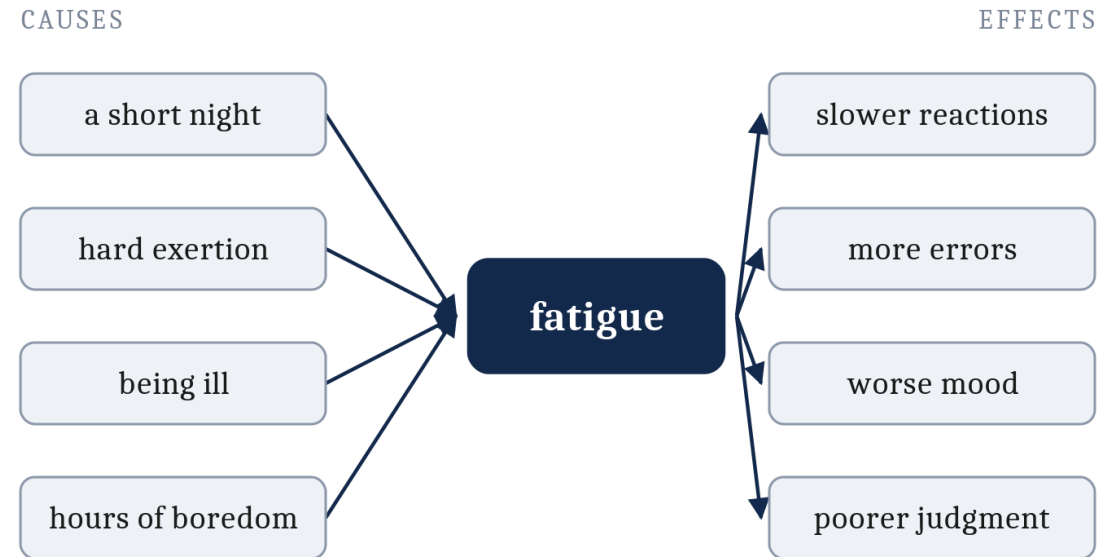
Four causes, four effects, tracked separately

sixteen connections to learn



One middle term, named “fatigue”

eight — and it predicts pairs nobody tested



The Whole Argument, and What to Carry

**Carry the three questions.
Let the roster of positions go.**

The habit that matters: when you meet a claim about the mind, ask *which level is this working at?*

**Next: Module 2 —
Comparative Approaches.** If a mind is a matter of organization, then minds elsewhere — other species, other machines — become a real question. That is where we go.

Mind is a real property of the physical — not a separate stuff, and not a redundant name for the brain



So it cannot be fully understood at any one level



So a science of mind needs a framework of levels



Marr's three: computational, algorithmic, implementational



Which line up with the disciplines, and with the evolutionary spine beneath them



So the course is built around them

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